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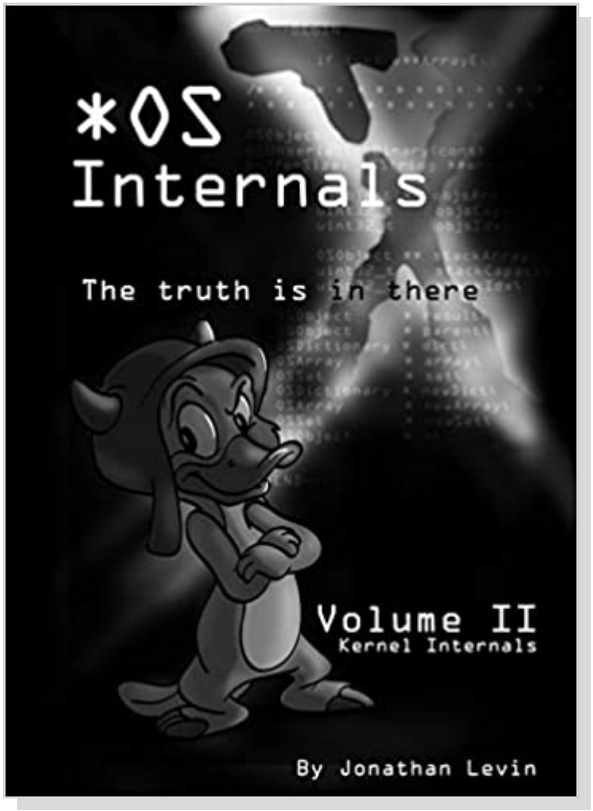
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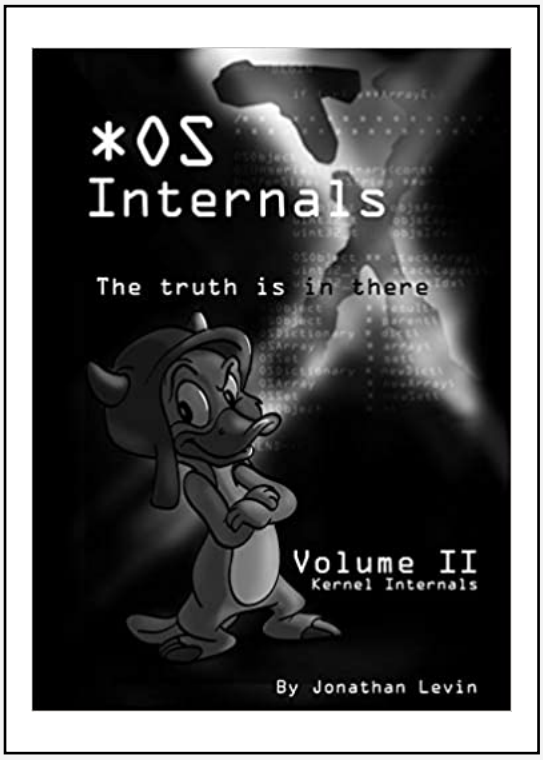
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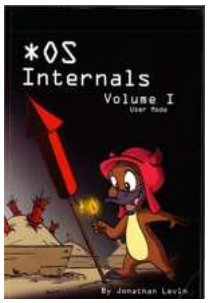
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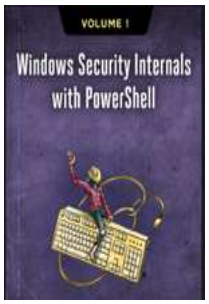
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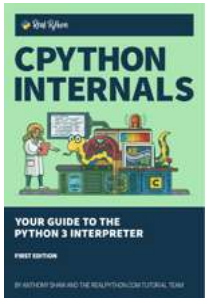
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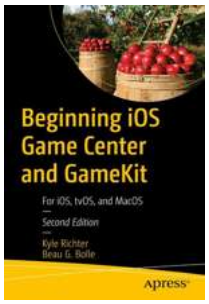
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*OS Internals

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Volume II

Kernel Mode

By Jonathan Levin

*OS Internals

(Mac OS X & iOS Internals, 2nd Edition)

Volume II: Kernel & Hardware

By Jonathan Levin

North Castle, NY

*OS Internals - Volume II - Kernel & Hardware

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Jonathan Levin
Phone: (617)-3000-667
Email: inquiries@NewOSXBook.com
Web: www.NewOSXBook.com

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About the [Author/Editor/Formatter/Publisher]

I started in my teens by hacking (mostly in the good sense of the word) and trying to figure out how things worked back in 1993, from an XT with a 2400 baud modem onto a shell I'm not even sure was mine. With no help per se, I had to `man man` and then `man *`..

A lot has happened in the two decades since, and from UNIX to Linux, I got to Windows and OS X. Over the years, I took up consulting and training - initially, in security. I then realized security is largely a projection of internals. Finally, I gathered a few good men and started Technologeeks.com - which is where I presently pass my time and function as CTO.

I'm still adjusting to authoring: I first took on Apple's OSes with "Mac OS X and iOS Internals" *(Wiley, 2012)*, which was well received. It was a painful process, but the dopamine rush was awesome. Now an addict, I took on Android, and that's when I also dared self-publishing for the first time, at the advice of my good friend Ronnie Federbush. That worked well - and gave me the chance to rewrite MOXiI the way I should have at first - free from publisher whims, free from censorship, and with no regard to page count or budget. Hence, you hold in your hand the second (but, technically, the final) volume of the new "*OS Internals" - a trilogy.

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Acknowledgments

MOXiI would never have been possible without the efforts of jailbreakers, which toil voluntarily and without compensation yet with incredulous efforts - all with the sole purpose of unrestricting Apple's amazing operating systems so that they mete out their full potential. I have had the honor of getting to know quite a few of them in person, and am deeply thankful to them all, including in particular:

- **@windknown** - Though unknown to most, surely all would recognize his group - Pangu! Hao and I forged a great friendship ever since we met for dinner in my favorite sushi joint in Sanlitun (Hatsune). He was also my first "surprise guest" in TechnoloGeeks' training, coming to NY to explain the inner workings of Pangu 9 for the very first time. Hao also helped me experiment - as with the other volumes - on a fully jailbroken iOS (12.1) for this book as well. 非常感谢, 我的朋友!
- **@key2fr** - a.k.a Ramtin Amin - known to the world as "the guy who made that lightning to serial cable". I had the honor of getting the very first cable he made! I later met him in Paris, and now have another reason to re-visit my favorite city every time. And that cable of his? That's just the tip of the iceberg. He has since demonstrated PCIe and NVMe wizardry which enabled, for the first time, the dumping of the SecureROM. What Ramtin can do to an iDevice over custom boards, soldering and JTAG is best left unspoken. Everything I know about the iDevice hardware, Ramtin taught me.
- **@_bazad** - a rising star in the world of *OS, this young researcher's first claim to fame was his "blanket" exploit, which demonstrated superior mastery of Mach ports providing a user-space jailbreak for iOS 11.3. I was truly honored when he presented this fantastic exploit in a world premiere as a mystery guest in @Technologeeks' training. This, however, was only a prelude to the "voucher_swap" exploit (now as a member of Google's Project Zero), which provided the core of the full iOS 12.1.2 jailbreak. Brandon didn't stop there - he also demonstrated an A12 PAC bypass, and provided important open source contributions, most notably `memctl`. Bran the Breaker reviewed the most difficult chapters of this work - those dealing with low level assembly, and the Mach primitives - and for that I am most grateful.
- **@s1guza** - already deserves my gratitude for being an admin of the NewOSXBook.com forum, and for taking my suggestions for his first public jailbreak, Phœnix. Since then, he has continued to break iOS time and time again, while producing unprecedented detail on Apple's most obscure protection mechanisms, notably APRR. This paragon hacker meticulously reviewed several chapters of this work.
- **@iH8sn0w** - a legendary jailbreaker who has made history on more than one occasion, thanks to his unparalleled mastery of iBoot. 'sn0w was willing to help me with helpful insights on all things iBoot and hardware.
- **@Xerub** - An anonymous but prolific hacker, who picked up where Sn0w left, crushing the last bits of iBoot and extending into SEP. Whatever details I have managed to glean of the latter would not have been remotely possible if not for his sharing of the SEP encrypted firmware keys. Xerub has also contributed to opensourcing jailbreaks (maybe overly so ;-), and will always be known as the one who brought all the gang together in 0x41 con.

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